

A Practical Guide to Dropout regularization

TOPIC -- DROPOUT REGULARIZATION

$$\tilde{h} = r \odot h, \quad r_i \sim \text{Bernoulli}(p)$$

-- 01 --

Early stopping can be viewed as regularization in time. Intuitively, a training procedure such as gradient descent tends to learn more and more complex functions with increasing iterations. By regularizing for time, model complexity can be controlled, improving generalization. Early stopping is implemented using one data set for training, one statistically independent data set for validation and another for testing. The model is trained until performance on the validation set no longer improves and then applied to the test set.

-- 02 --

$$w_0 = 0 \quad w_{t+1} = \left(I - \gamma n X^T X \right) w_t + \gamma n X^T Y \quad \{\displaystyle \begin{aligned} w_0 &= 0 \\ w_{t+1} &= \left(I - \frac{\gamma}{n} X^T X \right) \hat{w}_t + \frac{\gamma}{n} X^T Y \end{aligned} \}$$

-- 03 --

$$R(w) = \inf \left\{ \sum_{g=1}^G \|w_g\|_2^2 : w = \sum_{g=1}^G \bar{w}_g \right\} \quad \{\displaystyle R(w) = \inf \left\{ \sum_{g=1}^G \|w_g\|_2^2 : w = \sum_{g=1}^G \bar{w}_g \right\}$$

-- 04 --

Elastic net regularization tends to have a grouping effect, where correlated input features are assigned equal weights. Elastic net regularization is commonly used in practice and is implemented in many machine learning libraries.

-- 05 --

$$\text{prox}_{\lambda, R, g}(w_g) = \begin{cases} (1 - \lambda \|w_g\|_2) w_g, & \text{if } \|w_g\|_2 > \lambda \\ 0, & \text{if } \|w_g\|_2 \leq \lambda \end{cases} \quad \{\displaystyle \text{prox}_{\lambda, R, g}(w_g) = \begin{cases} (1 - \frac{\lambda}{\|w_g\|_2}) w_g, & \text{if } \|w_g\|_2 > \lambda \\ 0, & \text{if } \|w_g\|_2 \leq \lambda \end{cases} \}$$

-- 06 --

$$\min_{f_u \in \mathbb{R}^u} f^T L f = \min_{f_u \in \mathbb{R}^u} \{ f_u^T L u u^T f_u + f^T L L^T L u f_u + f_u^T L u f^T \}$$

$$\{\displaystyle \min_{f_u \in \mathbb{R}^u} f^T L f = \min_{f_u \in \mathbb{R}^u} \left\{ f_u^T L u u^T f_u + f^T L L^T L u f_u + f_u^T L u f^T \right\}$$

-- 07 --

The exact solution to the unregularized least squares learning problem minimizes the empirical error, but may fail. By limiting T , the only free parameter in the algorithm above, the problem is regularized for time, which may improve its generalization. The algorithm above is equivalent to restricting the number of gradient descent iterations for the empirical risk

-- 08 --

This regularizer constrains the functions learned for each task to be similar to the overall average of the functions across all tasks. This is useful for expressing prior information that each task is expected to share with each other task. An example is predicting blood iron levels measured at different times of the day, where each task represents an individual.

-- 09 --

This can be viewed as inducing a regularizer over the L_2 norm over members of each group followed by an L_1 norm over groups. This can be solved by the proximal method, where the proximal operator is a block-wise soft-thresholding function:

-- 10 --

where V is an underlying loss function that describes the cost of predicting $f(x)$ when the label is y , such as the square loss or hinge loss; and λ is a parameter which controls the importance of the regularization term. $R(f)$ is typically chosen to impose a penalty on the complexity of f . Concrete notions of complexity used include restrictions for smoothness and bounds on the vector space norm. A theoretical justification for regularization is that it attempts to impose Occam's razor on the solution (as depicted in the figure above, where the green function, the simpler one, may be preferred). From a Bayesian point of view, many regularization techniques correspond to imposing certain prior distributions on model parameters. Regularization can serve multiple purposes, including learning simpler models, inducing models to be sparse and introducing group structure into the learning problem. The same

idea arose in many fields of science. A simple form of regularization applied to integral equations (Tikhonov regularization) is essentially a trade-off between fitting the data and reducing a norm of the solution. More recently, non-linear regularization methods, including total variation regularization, have become popular.

-- 11 --

Other uses of regularization in statistics and machine learning Bayesian learning methods make use of a prior probability that (usually) gives lower probability to more complex models. Well-known model selection techniques include the Akaike information criterion (AIC), minimum description length (MDL), and the Bayesian information criterion (BIC). Alternative methods of controlling overfitting not involving regularization include cross-validation. Examples of applications of different methods of regularization to the linear model are:

-- 12 --

$$R(f_1 \cdots f_T) = \sum_{t=1}^T \|f_t - 1\|^2 + \sum_{s=1}^T \|f_s\|^2 + \lambda \sum_{k=1}^K \|f_k\|^2 \quad \{\displaystyle R(f_1 \cdots f_T) = \sum_{t=1}^T \left\| f_t - \frac{1}{T} \sum_{s=1}^T f_s \right\|^2 + \lambda \sum_{k=1}^K \|f_k\|^2\}$$

-- 13 --

where X and Y are the domains of input data x and their labels y respectively. Typically in learning problems, only a subset of input data and labels are available, measured with some noise. Therefore, the expected error is unmeasurable, and the best surrogate available is the empirical error over the N available samples:

-- 14 --

Regularizers for sparsity Assume that a dictionary ϕ_j with dimension p is given such that a function in the function space can be expressed as:

-- 15 --

Proximal methods While the L_1 norm does not result in an NP-hard problem, the L_1 norm is convex but is not strictly differentiable due to the kink at $x = 0$. Subgradient methods which rely on the subderivative can be used to solve L_1 regularized learning problems. However, faster convergence can be achieved through proximal methods. For a problem $\min_{w \in H} F(w) + R(w)$

$H\}F(w)+R(w)\}$ such that F $\{\displaystyle F\}$ is convex, continuous, differentiable, with Lipschitz continuous gradient (such as the least squares loss function), and R $\{\displaystyle R\}$ is convex, continuous, and proper, then the proximal method to solve the problem is as follows. First define the proximal operator